

The Hidden Cost of Delay

Analysis of 96.461 orders Olist · 2016–2018

Day 6

evaluation collapse
threshold

2x

average delivery time:
North vs. Southeast

48 days

range between the
best and worst sellers
(MG→RJ)

6,66%

delivered delayed orders

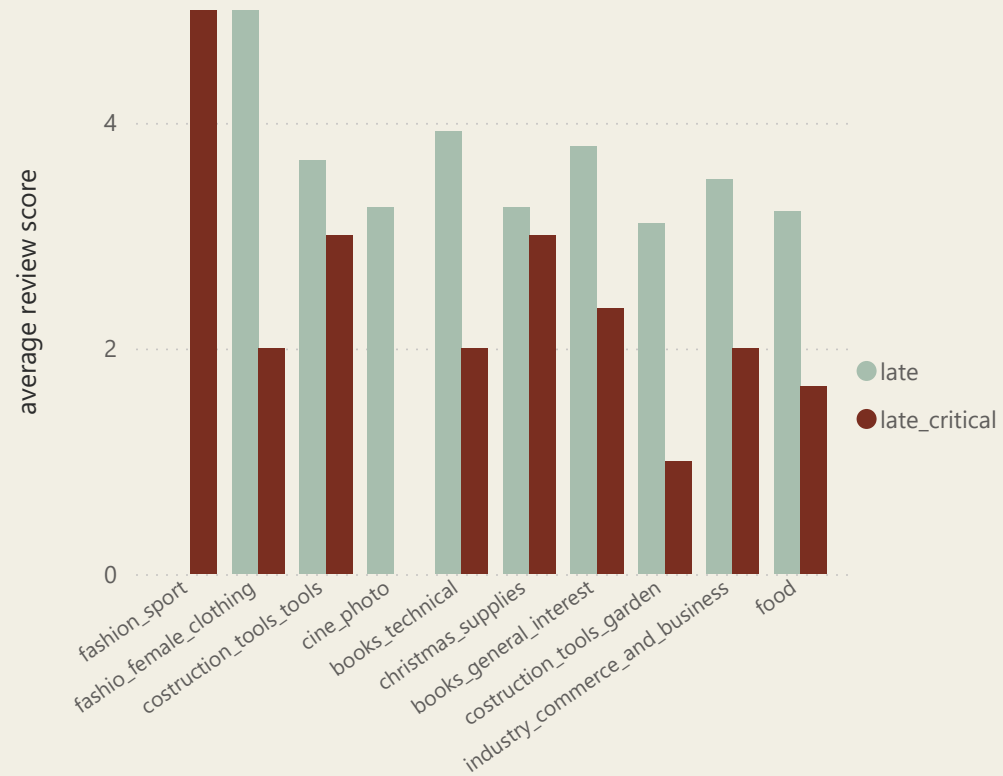
Source - public dataset Olist (Kaggle)

Period - 2016 to 2018

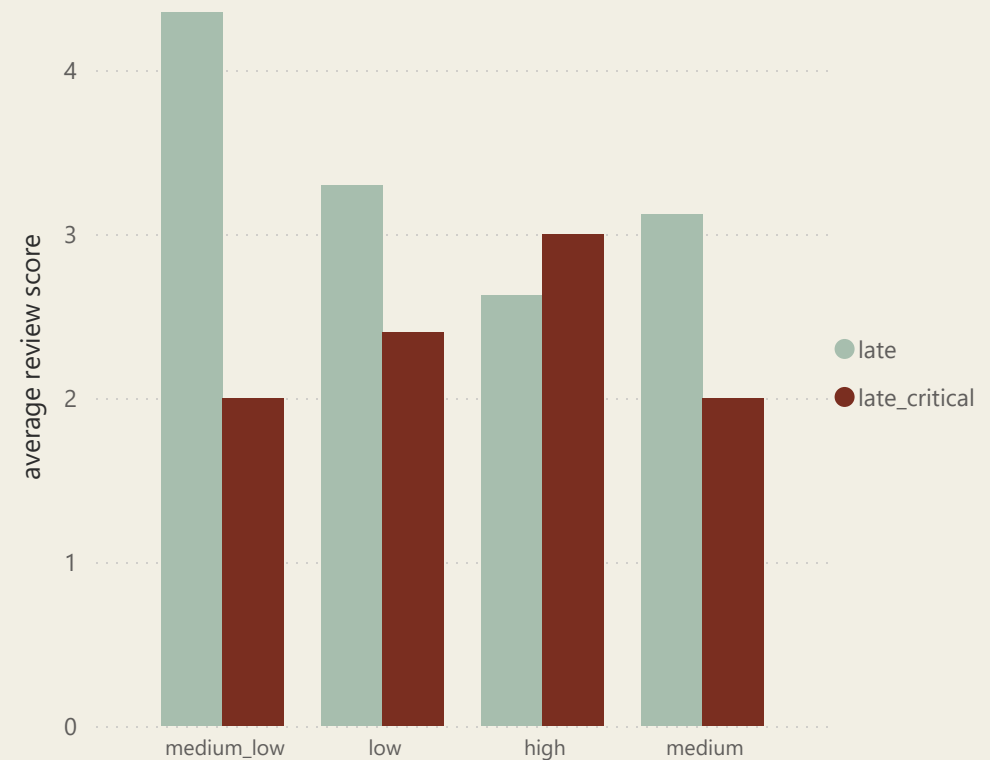
Analysed tables - orders, order_items, order_reviews, sellers, geolocation

The delay does not affect all contexts in the same way

Average rating by category — minor vs. critical delay

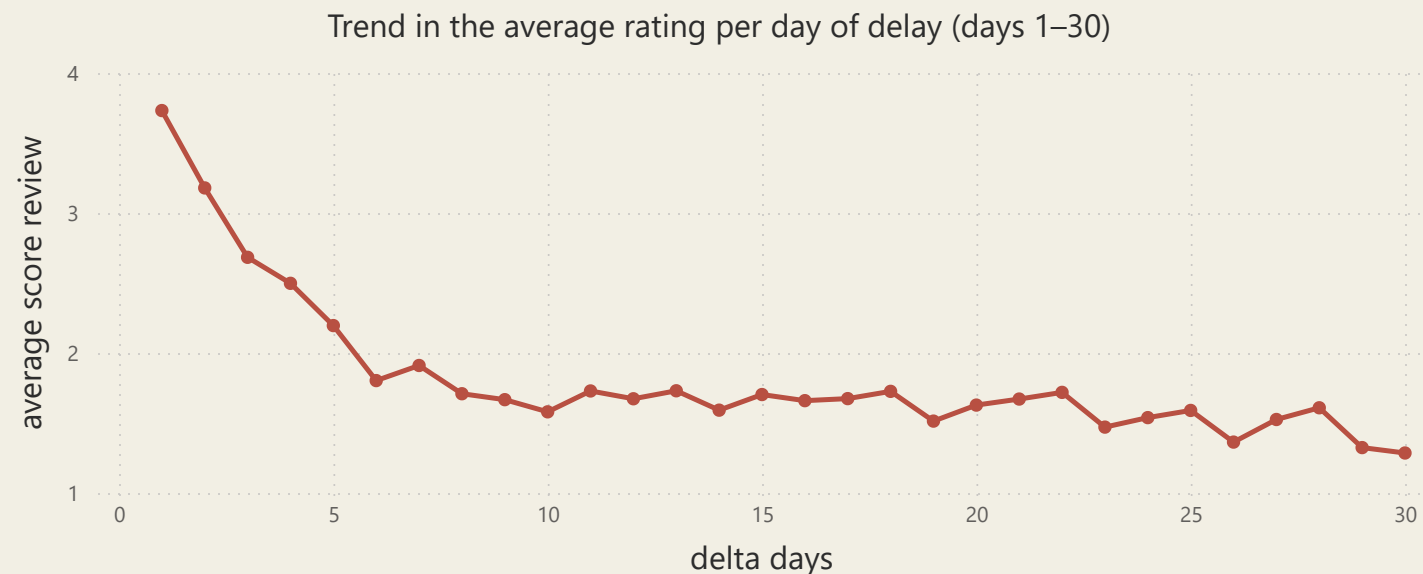


Average rating value tier — minor vs. critical delay



- The window for differentiation between categories is narrow: only during the first 5 days.
- The premium tier achieves an 87% score of 1 in late_critical—the most extreme result of the analysis.

There is a threshold — not a gradual decline



13% -> 69%

Score 1 proportion - Day 1 vs Day 6

3,73 -> 1,80

Average review Score - day 1 vs day 6

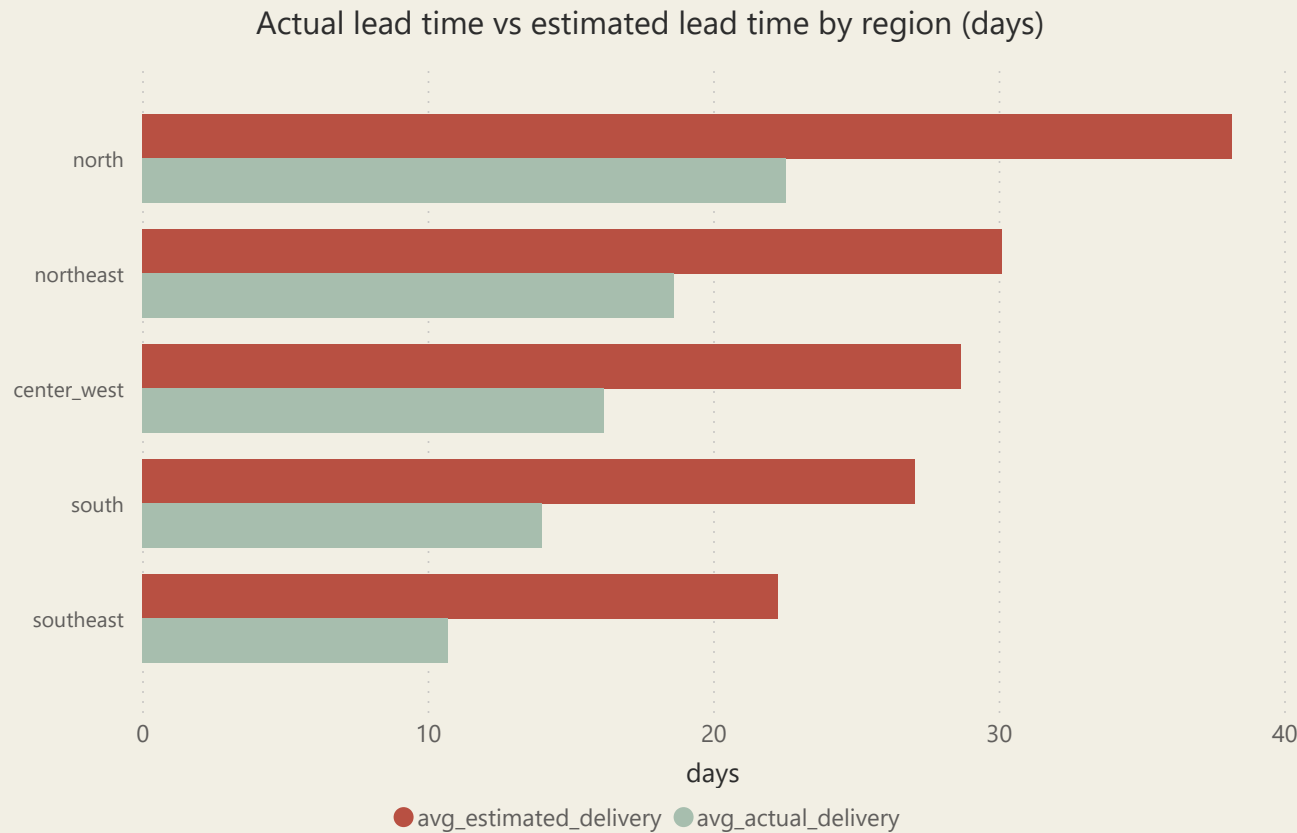
Up to 5 days, the customer still distinguishes between different levels of delay.

From the 6th day onwards, the score drops to around 1.8 and stabilises — 6 days or 30 days result in the same level of dissatisfaction

6

Day 6 —
threshold
(Mann-Whitney,
 $p < 0.05$)

The North is no longer falling behind — it simply takes longer, and is given more leeway to do so



10,69

South-East: 10.7 actual days · buffer of 11.6 days (ratio 1.08x)

18,63

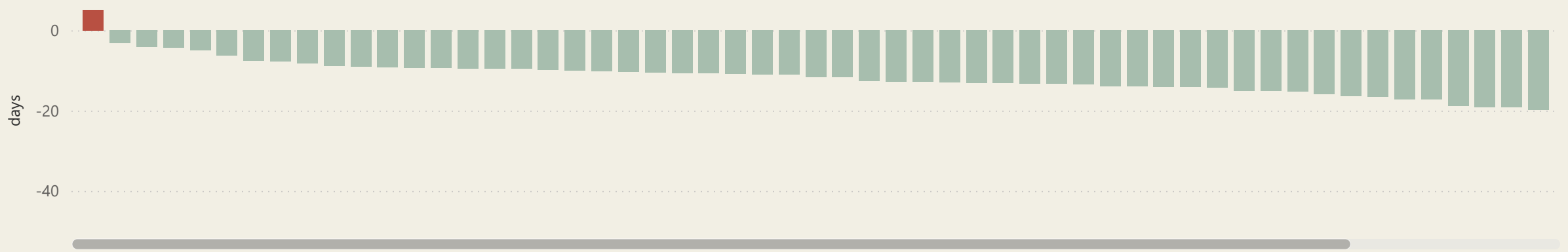
North-East: 18.6 actual days · margin 11.5 days (ratio 0.62x) — insufficient calibration

The problem is not the promise — it is the infrastructure.

But the North-East faces a double disadvantage: slow logistics and a disproportionately narrow margin of protection.

Same corridor, opposite results — the salesperson makes all the difference

Performance variation by salesperson — MG to RJ route (64 salespeople, 998 orders)
Kruskal-Wallis - $p < 0,0001$



Comparison: best vs worst seller on the MG > RJ route

Seller	Corridor	Orders	Average Late (%)	Average Delta	Average Actual	Average Std Actual	Average Std Delta
7ea5bfa6c340f58f8e71fc1f0412b0d6	MG_RJ	11	36,36%	5,1	30,1	38,31	38,53
b92e3c8f9738272ff7c59e111e108d7c	MG_RJ	10	0,00%	-43,1	14,7	4,99	22,89

Kruskal-Wallis $H = (p < 0.0001)$.

The difference between the best and worst sellers is statistically robust, with only 10–11 orders each — the actual range is far too large to be attributed to noise.

Where to intervene — and why

Take action before the 6th

Automatic alerts to the seller within the first 48 hours of a delay

Prioritise sensitive categories

Expectation management tailored to each category (toys, sports_leisure)

Rescheduling deadlines in the North-East

Adjust the `estimated\_delivery\_date` in proportion to the regional `avg\_actual`

Identify key salespeople

Process audit of sales staff with $\text{pct_late} > 30\%$ and $\text{avg_delta} > 0$